FULL GL2 DIFFERENCE GALOIS GROUP FOR ALGEBRAIC DISCRETE LAMÉ EQUATIONS

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Editorial note. The counted arithmetic Lamé theorem uses the sufficient GROUP-intersection condition printed at1272 and proved at1411–1426: Zh intersect(Zz0+Lambda)={0}. The universal coset equality printed1271 is not literally valid for cosets that do not contain0. The source-local arguments1317/1372/1377 require only absence of nonzero multiples ofh in those cosets, supplied by the explicitly proved group condition. Original formulas are retained; the unsupported universal coset equality is not certified.

Editorial note. For theta_k(z)=theta(z/k), the multiplier in550 has the divided weight exponent omega/k, as already printed in478,570–571 and1126/1134. The corresponding degree-zero periodicity wording in554 requires omega in kZ for the chosen representative. The author display/footnote are retained and their index slips are disclosed here.

Editorial note. The normalization Lambda=Z+tau Z belongs only to the generic analytic difference-field calculation. The algebraicity hypotheses on g2,g3,h,A,B are read in the ORIGINAL Weierstrass coordinates of76–110 and the arithmetic proof1414–1426. A period normalization need not preserve these algebraic values. The generic theorem allows complex coefficients, and coordinate rescaling is understood through the usual difference-field identification; no simultaneous algebraicity assumption in the normalized coordinates is imposed.

Editorial note. The cover arrow in419–420 is contravariant to the printed field inclusion418 and has its direction determined by405–408. At941 the required ambient field is the already defined M_(Lambda-double-prime) from916–940; the stronger printed M_(kLambda) membership is not endorsed for arbitrary sublattice pairs solely from their indexk. The two actual applications use the explicit fields at999–1005 and1057–1065. Original equations and field labels are retained with this scope explanation.

Editorial note. The determinant step1403–1406 is intentionally abbreviated by the author. It uses the already allowed standard first-order multiplicative PV criterion together with the printed pole divisor ofb and the proved nonresonance condition. Independent review found these sufficient. The source abbreviation is retained; the reviewer verification is not inserted as newly authored proof or counted separately.

Allowed standard prerequisites (editorial).

- Picard–Vessiot difference Galois theory, the propertyP structural theorem, and first-order determinant characters. The standard first-order determinant-character criterion says that finite image implies b^m=phi(f)/f for some integerm>=1 andf inK*. This general criterion is allowed; its application to the actual elliptic divisor/nonresonance data remains the local proof obligation, not a supplied infinite-image conclusion.. General algebraic-group interpretation only; author proof that the elliptic union field satisfiesP remains required.

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- Tsen, curve/function-field duality, Riemann–Hurwitz and classification of morphisms of complex elliptic curves. Classical tools used inside retained finite-translation-extension argument.
- Hilbert90 for a finite cyclic extension. The author cocycle normalization and two-stage Riccati descent remain required.
- Jacobi theta quasiperiodicity, theta product representation and elementary Weierstrass divisor facts. Source conventions retained; bespoke Riccati theta-gcd and weight constraints not assumed.
- Generic second-order imprimitive group criterion via iterated Riccati equation and algebraic subgroups ofGL2. Generic difference-field input; actual elliptic-field reduction and nonresonant pole analysis remain proved.
- Schneider elliptic transcendence theorem. Standard final arithmetic input, not counted as newly proved here.

1. Introduction

Let $\mathcal{E} \subset \mathbb{P}^2$ be the elliptic curve defined by the projectivization of the Weierstrass equation

$$y^2 = 4x^3 - g_2 x - g_3 \qquad \text{with} \quad g_2, g_3 \in \mathbb{C}. \tag{1.1}$$

We denote by $\mathcal{E}(\mathbb{C})$ the group of $\mathbb{C}$-points of $\mathcal{E}$. Its abelian group law is denoted by $\oplus$.

In this paper, we study the difference Galois groups of linear difference equations of order two on $\mathcal{E}(\mathbb{C})$ of the form:

$$\underline{y}(\underline{z} \oplus 2\underline{h}) + \underline{a}(\underline{z})\underline{y}(\underline{z} \oplus \underline{h}) + \underline{b}(\underline{z})\underline{y}(\underline{z}) = 0, \tag{1.2}$$

where $\underline{y}$ is an unknown function of the variable $\underline{z} \in \mathcal{E}(\mathbb{C})$, $\underline{h}$ is a fixed non torsion point of $\mathcal{E}(\mathbb{C})$ and $\underline{a}$, $\underline{b}$ are given rational functions on $\mathcal{E}$.

This equation can be seen as a difference equation over $\mathbb{C}$. Indeed, if $\Lambda \subset \mathbb{C}$ is a lattice of periods of $\mathcal{E}$ and if $\wp$ is the corresponding Weierstrass function, then $\mathbb{C}/\Lambda$ is identified with $\mathcal{E}(\mathbb{C})$ via the factorization through $\mathbb{C}/\Lambda$ of

$$\varphi : \ z \in \mathbb{C} \mapsto (\wp(z) : \wp'(z) : 1) \in \mathcal{E}(\mathbb{C}).$$

Pulling back the equation (1.2) via φ, which is a group morphism from $(\mathbb{C}, +)$ to $(\mathcal{E}(\mathbb{C}), \oplus)$, we obtain the following difference equation on $\mathbb{C}$:

$$y(z + 2h) + a(z)y(z + h) + b(z)y(z) = 0, \tag{1.3}$$

where y is an unknown function of the variable $z \in \mathbb{C}$, $a := \underline{a} \circ \varphi$ and $b := \underline{b} \circ \varphi$ are Λ-periodic elliptic functions and $h \in \varphi^{-1}(\underline{h})$. So, we have the following relations between the equations (1.2) and (1.3): $\underline{z} = \varphi(z)$, $\underline{h} = \varphi(h)$ and $\underline{y}(\underline{z}) = y(z)$.

2. Difference Galois theory: reminders and complements

2.1. **Generalities on difference Galois theory.** For details on what follows, we refer to [13, Chapter 1].

A difference ring (R, ϕ) is a ring R together with a ring automorphism $\phi : R \to R$. An ideal of R stabilized by ϕ is called a difference ideal of (R, ϕ). If R is a field then (R, ϕ) is called a difference field.

The ring of constants R^ϕ of the difference ring (R, ϕ) is defined by

$$R^\phi := \{f \in R \,|\, \phi(f) = f\}.$$

A difference ring morphism (resp. difference ring isomorphism) from the difference ring (R, ϕ) to the difference ring $(\widetilde{R}, \widetilde{\phi})$ is a ring morphism (resp. ring isomorphism) $\varphi : R \to \widetilde{R}$ such that $\varphi \circ \phi = \widetilde{\phi} \circ \varphi$.

A difference ring $(\widetilde{R},\widetilde{\phi})$ is a difference ring extension of a difference ring (R,ϕ) if $\widetilde{R}$ is a ring extension of R and $\widetilde{\phi}_{|R}=\phi$; in this case, we will often denote $\widetilde{\phi}$ by ϕ. Two difference ring extensions $(\widetilde{R}_1,\widetilde{\phi}_1)$ and $(\widetilde{R}_2,\widetilde{\phi}_2)$ of a difference ring (R,ϕ) are isomorphic over (R,ϕ) if there exists a difference ring isomorphism φ from $(\widetilde{R}_1,\widetilde{\phi}_1)$ to $(\widetilde{R}_2,\widetilde{\phi}_2)$ such that $\varphi_{|R}=\mathrm{Id}_R$.

We now let (K,ϕ) be a difference field. We assume that its field of constants $C:=K^\phi$ is algebraically closed and that the characteristic of K is 0.

Consider a linear difference system

$$\phi Y = AY \qquad \text{with} \quad A\in \mathrm{GL}_n(K). \tag{2.1}$$

According to [13, Section 1.1], there exists a difference ring extension (R,ϕ) of (K,ϕ) such that

1) there exists $U\in \mathrm{GL}_n(R)$ such that $\phi(U)=AU$ (such a U is called a fundamental matrix of solutions of (2.1));
2) R is generated, as a K-algebra, by the entries of U and $\det(U)^{-1}$;
3) the only difference ideals of (R,ϕ) are $\{0\}$ and R.

Such a difference ring (R,ϕ) is called a Picard–Vessiot ring for (2.1) over (K,ϕ). It is unique up to isomorphism of difference rings over (K,ϕ). It is worth mentioning that $R^\phi=C$; see [13, Lemma 1.8].

Remark 2.1. Picard–Vessiot rings are not domains in general: they are finite direct sums of domains cyclically permuted by ϕ; see [13, Corollary 1.16].

The corresponding difference Galois group G over (K,ϕ) of (2.1) is the group of K-linear ring automorphisms of R commuting with ϕ:

$$G:=\{\sigma\in \mathrm{Aut}(R/K)\,|\,\phi\circ\sigma=\sigma\circ\phi\}.$$

The choice of the base field is by no way innocent. The bigger the base field is, the smaller the Galois group is.

A straightforward computation shows that, for any $\sigma\in G$, there exists a unique $C(\sigma)\in \mathrm{GL}_n(C)$ such that $\sigma(U)=UC(\sigma)$. According to [13, Theorem 1.13], one can identify G with an *algebraic* subgroup of $\mathrm{GL}_n(C)$ via the faithful representation

$$\sigma\in G\mapsto C(\sigma)\in \mathrm{GL}_n(C).$$

If we choose another fundamental matrix of solutions U, we find a conjugate representation.

Remark 2.2. Given an nth order difference equation

$$a_n\phi^n(y)+\cdots+a_1\phi(y)+a_0y=0, \tag{2.2}$$

with $a_0,\dots,a_n\in K$ and $a_0a_n\neq 0$, we can consider the equivalent linear difference system

$$\phi Y=AY, \qquad \text{with} \quad A=\begin{pmatrix} 0 & 1 & 0 & \dots & 0\\ 0 & 0 & 1 & \ddots & \vdots\\ \vdots & \vdots & \ddots & \ddots & 0\\ 0 & 0 & \dots & 0 & 1\\ -\frac{a_0}{a_n} & -\frac{a_1}{a_n} & \dots & \dots & -\frac{a_{n-1}}{a_n}\end{pmatrix}\in \mathrm{GL}_n(K). \tag{2.3}$$

By "Galois group of the difference equation (2.2)" we mean "Galois group of the difference system (2.3)".

We shall now introduce a property relative to the difference base field, which appears in [13, Lemma 1.19].

Definition 2.3. We say that the difference field (K,ϕ) satisfies property $(\mathcal{P})$ if the following properties hold:

- the field K is a $\mathcal{C}^1$-field[1];
- if L is a finite field extension of K such that ϕ extends to a field endomorphism of L then $L=K$.

Theorem 2.4. *Assume that (K,ϕ) satisfies property $(\mathcal{P})$. Let $K^\phi = C$. Let $G\subset \mathrm{GL}_n(C)$ be the difference Galois group over (K,ϕ) of*

$$\phi(Y)=AY, \qquad \text{with} \quad A\in \mathrm{GL}_n(K). \tag{2.4}$$

Then, the following properties hold:

- *G/G° is cyclic, where G° is the identity component of G;*
- *there exists $B\in G(K)$ such that (2.4) is isomorphic to $\phi Y = BY$ over K.*

Let $\widetilde{G}$ be an algebraic subgroup of $\mathrm{GL}_n(C)$ such that $A\in\widetilde{G}(K)$. The following properties hold:

- *G is conjugate to a subgroup of $\widetilde{G}$;*
- *any minimal element in the set of algebraic subgroups $\widetilde{H}$ of $\widetilde{G}$ for which there exists $T\in\mathrm{GL}_n(K)$ such that $\phi(T)AT^{-1}\in\widetilde{H}(K)$ is conjugate to G;*
- *G is conjugate to $\widetilde{G}$ if and only if, for any $T\in\widetilde{G}(K)$ and for any proper algebraic subgroup $\widetilde{H}$ of $\widetilde{G}$, one has that $\phi(T)AT^{-1}\notin\widetilde{H}(K)$.*

Proof. The proof of [13, Propositions 1.20 and 1.21] in the special case where $K:=C(z)$ and ϕ is the shift $\phi(f(z)):=f(z+h)$ with $h\in C^\times$, extends *mutatis mutandis* to the present case. □

2.2. **Difference equations on elliptic curves.** Let $\Lambda\subset\mathbb{C}$ be a lattice. Without loss of generality, we can assume that

$$\Lambda=\mathbb{Z}+\mathbb{Z}\tau, \qquad \text{with} \quad \Im(\tau)>0,$$

where $\Im(\cdot)$ denotes the imaginary part. For any lattice $\Lambda'\subset\mathbb{C}$, we let $\mathrm{M}_{\Lambda'}$ be the field of Λ'-periodic meromorphic functions. We denote by K the field defined by

$$K:=\bigcup_{\Lambda' \text{ sub-lattice of } \Lambda} \mathrm{M}_{\Lambda'} = \bigcup_{k\geq 1}\mathrm{M}_{k\Lambda}\,.$$

Let $h\in\mathbb{C}$ such that $h \mod \Lambda$ is not a torsion point of $\mathbb{C}/\Lambda$. We endow K with the non-cyclic field automorphism ϕ defined by

$$\phi(f)(z):=f(z+h).$$

Then, (K,ϕ) is a difference field.

Proposition 2.5. *The field of constants of (K,ϕ) is*

$$K^\phi=\mathbb{C}.$$

Proof. Consider $f\in K^\phi$. Let Λ' be a sub-lattice of Λ such that $f\in\mathrm{M}_{\Lambda'}$. Note that f is Λ'-periodic (because $f\in\mathrm{M}_{\Lambda'}$) and h-periodic (because $\phi(f)=f$), so f is a $(\Lambda'+h\mathbb{Z})$-periodic meromorphic function. But $\Lambda'+h\mathbb{Z}$ has an accumulation point because $h \mod \Lambda$ is not a torsion point of $\mathbb{C}/\Lambda$. Therefore, f is constant. □

Proposition 2.6. *The difference field (K,ϕ) satisfies property $(\mathcal{P})$ (see Definition 2.3).*

[1]Recall that K is a $\mathcal{C}^1$-field if every non-constant homogeneous polynomial P over K has a non-trivial zero provided that the number of its variables is more than its degree.

Proof. Since $K = \bigcup_{k\geq 1} \mathrm{M}_{k\Lambda}$ is the increasing union of the fields $\mathrm{M}_{k\Lambda}$, the fact that K is a $\mathcal{C}^1$-field follows from Tsen's theorem [6] (according to which the function field of any algebraic curve over an algebraically closed field, e.g., $\mathrm{M}_{k\Lambda}$, is $\mathcal{C}^1$).

Let L be a finite extension of K such that ϕ extends to a field endomorphism of L. We have to prove that $L = K$. The primitive element theorem ensures that there exists $u \in L$ such that $L = K(u)$. Let Λ' be a sub-lattice of Λ such that

- u is algebraic over $\mathrm{M}_{\Lambda'}$,
- $\phi(u) \in \mathrm{M}_{\Lambda'}(u)$.

Then, $\mathrm{M}_{\Lambda'}(u)$ is a finite extension of $\mathrm{M}_{\Lambda'}$ and ϕ induces an automorphism of $\mathrm{M}_{\Lambda'}(u)$.

Using the equivalence of categories between between smooth projective curves and function fields of dimension 1 [3, Corollary 6.12], we see that there exists a commutative diagram of the form

$$\begin{array}{ccc} X & \xrightarrow{f} & X \\ {\scriptstyle\varphi}\downarrow & & \downarrow{\scriptstyle\varphi} \\ \mathbb{C}/\Lambda' & \xrightarrow[z\mapsto z+h]{} & \mathbb{C}/\Lambda' \end{array}$$

where $\varphi : X \to \mathbb{C}/\Lambda'$ is a morphism of smooth projective curves, whose induced morphism of function fields "is" the inclusion $\mathrm{M}_{\Lambda'} \subset \mathrm{M}_{\Lambda'}(u)$, and where f is an endomorphism of X, whose induced morphism on function fields "is" $\phi : \mathrm{M}_{\Lambda'}(u) \to \mathrm{M}_{\Lambda'}(u)$. Considering this commutative diagram, we see that f has degree 1 and that, if φ is ramified above $y \in \mathbb{C}/\Lambda'$, then φ is also ramified above $y - h$. So, the set of ramification values of φ is stable by $z \mapsto z - h$. This set being finite, it has to be empty. So, φ is unramified.

Hurwitz's formula implies that X has genus 1, i.e., that X is an elliptic curve. So, there exist a lattice $\Lambda'' \subset \mathbb{C}$ and an isomorphism $\psi : \mathrm{M}_{\Lambda'}(u) \to \mathrm{M}_{\Lambda''}$. There exists $(a, b) \in \mathbb{C}^\times \times \mathbb{C}$ such that $a\Lambda'' \subset \Lambda'$ and such that the restriction $\psi_{|\,\mathrm{M}_{\Lambda'}}$ is given, for all $f \in \mathrm{M}_{\Lambda'}$, by $\psi(f)(z) = f(az + b)$. (Indeed, $\psi_{|\,\mathrm{M}_{\Lambda'}} : \mathrm{M}_{\Lambda'} \to \mathrm{M}_{\Lambda''}$ is a field morphism from the function field of the elliptic curve $\mathbb{C}/\Lambda'$ to the function field of the elliptic curve $\mathbb{C}/\Lambda''$. So, $\psi_{|\,\mathrm{M}_{\Lambda'}}$ is induced by a morphism from the elliptic curve $\mathbb{C}/\Lambda''$ to the elliptic curve $\mathbb{C}/\Lambda'$. Now, our claim follows from the fact that the morphisms from $\mathbb{C}/\Lambda''$ to $\mathbb{C}/\Lambda'$ are of the form $z \mod \Lambda'' \mapsto az + b \mod \Lambda'$ for some $(a, b) \in \mathbb{C}^\times \times \mathbb{C}$ such that $a\Lambda'' \subset \Lambda'$.) The commutative diagram

$$\begin{array}{ccccc} \mathrm{M}_{\Lambda'} & \hookrightarrow & \mathrm{M}_{\Lambda'}(u) & & \\ & {\scriptstyle v(z)\mapsto v(az+b)}\searrow & \downarrow{\scriptstyle\psi} & \swarrow & \\ & & \mathrm{M}_{\Lambda''} & \xrightarrow[w(z)\mapsto w(\frac{z-b}{a})]{} & \mathrm{M}_{a\Lambda''} \end{array}$$

shows that the fields $\mathrm{M}_{\Lambda'}(u)$ and $\mathrm{M}_{a\Lambda''}$ are $\mathrm{M}_{\Lambda'}$-isomorphic. But the extension $\mathrm{M}_{a\Lambda''} / \mathrm{M}_{\Lambda'}$ is Galois (indeed, this is equivalent to the fact that the corresponding morphism of smooth projective curves $\mathbb{C}/\Lambda' \to \mathbb{C}/a\Lambda''$ is Galois, and this is easily seen from the explicit description of the morphisms between these curves). Therefore, any $\mathrm{M}_{\Lambda'}$-morphism from $\mathrm{M}_{a\Lambda''}$ to $K(u)$ must leave $\mathrm{M}_{a\Lambda''}$ globally invariant. But, $\mathrm{M}_{a\Lambda''}$ and $\mathrm{M}_{\Lambda'}(u)$ are $\mathrm{M}_{\Lambda'}$-isomorphic subfields of $K(u)$. So $\mathrm{M}_{\Lambda'}(u) \subset \mathrm{M}_{a\Lambda''}$, and therefore $u \in \mathrm{M}_{a\Lambda''} \subset K$ and $L = K(u) \subset K$. □

Corollary 2.7. *The conclusions of Theorem* 2.4 *are valid for* (K, ϕ).

3. Theta functions and Weierstrass $\wp$-function

3.1. **Theta functions.** We recall that

$$\Lambda = \mathbb{Z} + \tau\mathbb{Z} \subset \mathbb{C} \qquad \text{with} \quad \Im(\tau) > 0.$$

Let θ be the Jacobi theta function defined by

$$\theta(z) = \sum_{m\in\mathbb{Z}} (-1)^m e^{i\pi m(m-1)\tau} e^{2i\pi mz}.$$

We shall now recall some basic facts about this function. We refer to [8, Chapter I] for details and proofs.

Remark 3.1. The classical theta function is defined by $\vartheta(z,\tau) = \sum_{m\in\mathbb{Z}} e^{i\pi m^2\tau + 2i\pi mz}$. Actually, this is the function studied in [8, Chapter I]. But, there is a simple relation between θ and ϑ, namely $\theta(z) = \vartheta(z + \frac{1-\tau}{2}, \tau)$. So that any statement for ϑ can be immediately translated into a statement for θ.

We recall that θ is a 1-periodic entire function such that

$$\theta(z+\tau) = -e^{-2i\pi z}\theta(z).$$

Moreover, we have the following formula, known as Jacobi's triple product formula:

$$\theta(z) = \prod_{m=1}^{\infty} \big(1 - e^{2i\pi\tau m}\big)\big(1 - e^{2i\pi((m-1)\tau + z)}\big)\big(1 - e^{2i\pi(m\tau - z)}\big).$$

For any integer $k \geq 1$, we let θ_k be the function given by

$$\theta_k(z) := \theta(z/k).$$

This k-periodic entire function satisfies the following functional equation:

$$\theta_k(z + k\tau) = -e^{-2i\pi z/k}\theta_k(z). \tag{3.1}$$

It follows from Jacobi's triple product formula that the zeroes of θ_k are simple and that its set of zeroes is $k\Lambda$.

Let Θ_k be the set of entire functions of the form

$$c\prod_{\xi\in\mathbb{C}} \theta_k(z-\xi)^{n_\xi}$$

with $c \in \mathbb{C}^\times$ and $(n_\xi)_{\xi\in\mathbb{C}} \in \mathbb{N}^{(\mathbb{C})}$ with finite support. We denote by Θ_k^{quot} the set of meromorphic functions over $\mathbb{C}$ that can be written as quotient of two elements of Θ_k.

We define the divisor $\mathrm{div}_k(f)$ of $f \in \Theta_k^{\mathrm{quot}}$ as the following formal sum of points of $\mathbb{C}/k\Lambda$:

$$\mathrm{div}_k(f) := \sum_{\lambda\in\mathbb{C}/k\Lambda} \mathrm{ord}_\lambda(f)[\lambda],$$

where $\mathrm{ord}_\lambda(f)$ is the $(z-\xi)$-adic valuation of f, for an arbitrary $\xi \in \lambda$ (it does not depend on the chosen $\xi \in \lambda$). For any $\lambda \in \mathbb{C}/k\Lambda$ and any $\xi \in \lambda$, we set

$$[\xi]_k := [\lambda].$$

Moreover, we will write

$$\sum_{\lambda\in\mathbb{C}/k\Lambda} n_\lambda[\lambda] \leq \sum_{\lambda\in\mathbb{C}/k\Lambda} m_\lambda[\lambda]$$

if, for all $\lambda \in \mathbb{C}/k\Lambda$, $n_\lambda \leq m_\lambda$. We also introduce the weight $\omega_k(f)$ of f defined by

$$\omega_k(f) := \sum_{\lambda\in\mathbb{C}/k\Lambda} \mathrm{ord}_\lambda(f)\lambda \in \mathbb{C}/k\Lambda$$

and its degree $\deg_k(f)$ given by

$$\deg_k(f) := \sum_{\lambda \in \mathbb{C}/k\Lambda} \operatorname{ord}_\lambda(f) \in \mathbb{Z}.$$

If $f = c \prod_{\xi \in \mathbb{C}} \theta_k(z-\xi)^{n_\xi} \in \Theta_k^{\mathrm{quot}}$, then

$$\operatorname{div}_k(f) = \sum_{\xi \in \mathbb{C}} n_\xi [\xi]_k, \qquad \omega_k(f) = \sum_{\xi \in \mathbb{C}} n_\xi \xi \mod k\Lambda \qquad \text{and} \qquad \deg_k(f) = \sum_{\xi \in \mathbb{C}} n_\xi.$$

The interest of Θ_k^{quot} in our context is given by the following classical result.

Proposition 3.2. *We have*

$$\mathrm{M}_{k\Lambda}^\times \subset \Theta_k^{\mathrm{quot}}.$$

Proof. This inclusion means that any $k\Lambda$-periodic meromorphic function can be written, up to some multiplicative constant in $\mathbb{C}^\times$, as a quotient of product of functions of the form $\theta_k(z-\xi)$. This is classical, see [8, Chapter I, Section 6]. □

We now state a couple of lemmas, which will be used freely in the rest of the paper.

Lemma 3.3. *Any $f = c \prod_{\xi \in \mathbb{C}} \theta_k(z-\xi)^{n_\xi} \in \Theta_k^{\mathrm{quot}}$ is k-periodic and satisfies*

$$f(z+k\tau) = (-1)^{\deg_k(f)} e^{2i\pi\omega} e^{-2i\pi \deg_k(f) z/k} f(z), \tag{3.2}$$

where $\omega = \sum_{\xi \in \mathbb{C}} n_\xi \xi$ is a representative of $\omega_k(f)$[2]. Conversely, any non zero k-periodic meromorphic function f over $\mathbb{C}$ such that

$$f(z+k\tau) = c e^{-2i\pi n z/k} f(z), \tag{3.3}$$

for some $c \in \mathbb{C}^\times$ and $n \in \mathbb{Z}$, belongs to Θ_k^{quot}.

Proof. The fact that any $f \in \Theta_k^{\mathrm{quot}}$ is k-periodic and satisfies the functional equation (3.2) follows from the fact that θ_k is k-periodic and satisfies the functional equation (3.1). Conversely, consider a non zero k-periodic meromorphic function f over $\mathbb{C}$ satisfying an equation of the form (3.3). Using the functional equation (3.1), we see that the k-periodic meromorphic function $g(z) = \frac{f(z)\theta_k(z-\xi)}{\theta_k(z)^n \theta_k(z)}$, where $\xi \in \mathbb{C}$ is such that $e^{-2i\pi\xi/k} = (-1)^n c$, satisfies $g(z+k\tau) = g(z)$. So g belongs to $\mathrm{M}_{k\Lambda}^\times \subset \Theta_k^{\mathrm{quot}}$, whence the result. □

Lemma 3.4. *If $f \in \Theta_k$ is such that $\deg_k(f) = 0$ then f is constant.*

Proof. Consider $f \in \Theta_k$. There exists $c \in \mathbb{C}^\times$ and $(n_\xi)_{\xi \in \mathbb{C}} \in \mathbb{N}^{(\mathbb{C})}$ with finite support such that

$$f(z) = c \prod_{\xi \in \mathbb{C}} \theta_k(z-\xi)^{n_\xi}.$$

Then, $\deg_k(f) = \sum_{\xi \in \mathbb{C}} n_\xi$ is equal to 0 by hypothesis. Thus, for all $\xi \in \mathbb{C}$, $n_\xi = 0$ and hence $f = c$ is constant. □

[2]It follows from this formula that f belongs to $\mathrm{M}_{k\Lambda}$ if and only if $\deg_k(f) = \sum_{\xi \in \mathbb{C}} n_\xi = 0$ and $\omega = \sum_{\xi \in \mathbb{C}} n_\xi \xi \in \mathbb{Z}$.

3.2. **Weierstrass $\wp$-function.** For details on what follows, we refer to [12, Chapter VI]. Recall that

$$\wp(z) := \frac{1}{z^2} + \sum_{\lambda\in\Lambda\setminus\{0\}} \frac{1}{(z+\lambda)^2} - \frac{1}{\lambda^2} \in \mathrm{M}_\Lambda$$

denotes the Weierstrass elliptic function associated to the lattice Λ. For any integer $k \geq 1$, we denote by $\wp_k \in \mathrm{M}_{k\Lambda}$ the Weierstrass function defined by

$$\wp_k(z) := \wp(z/k) \in \mathrm{M}_{k\Lambda}\,.$$

This $k\Lambda$-periodic meromorphic function is an even function, its poles are of order two and its set of poles is $k\Lambda$. Therefore, its derivative $\wp_k'$ is an odd function, its poles are of order three and its set of poles is $k\Lambda$.

Any $k\Lambda$-periodic elliptic function is a rational function in $\wp_k$ and $\wp_k'$, that is

$$\mathrm{M}_{k\Lambda} = \mathbb{C}(\wp_k, \wp_k').$$

4. Irreducibility of the difference Galois group

We let

$$\phi^2(y) + a\phi(y) + by = 0 \qquad \text{with} \quad a \in \mathrm{M}_\Lambda \quad \text{and} \quad b \in \mathrm{M}_\Lambda^\times \tag{4.1}$$

be a difference equation of order 2 with coefficients in M_Λ and we denote by

$$\phi Y = AY \qquad \text{with} \quad A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix} \in \mathrm{GL}_2(\mathrm{M}_\Lambda)$$

the associated difference system. For the notations M_Λ, ϕ, K, etc, we refer to Sections 2 and 3.

We let $G \subset \mathrm{GL}_2(\mathbb{C})$ be the difference Galois group over (K, ϕ) of equation (4.1). According to Corollary 2.7, G is an algebraic subgroup of $\mathrm{GL}_2(\mathbb{C})$ such that the quotient G/G° of G by its identity component G° is cyclic. A direct inspection of the classification, up to conjugation, of the algebraic subgroups of $\mathrm{GL}_2(\mathbb{C})$ given in [9, Theorem 4] shows that G satisfies one of the following properties:

- The group G is reducible (i.e., conjugate to some subgroup of the group of upper-triangular matrices in $\mathrm{GL}_2(\mathbb{C})$). If G is reducible, we distinguish the following sub-cases:
  - the group G is completely reducible (i.e., is conjugate to some subgroup of the group of diagonal matrices in $\mathrm{GL}_2(\mathbb{C})$);
  - the group G is not completely reducible.
- The group G is irreducible (i.e., not reducible) and imprimitive (see Section 5 for the definition).
- The group G is irreducible and is not imprimitive, and, in this case, there exists an algebraic subgroup μ of $\mathbb{C}^\times$ such that $G = \mu\,\mathrm{SL}_2(\mathbb{C})$. Therefore, $G = \{M \in \mathrm{GL}_2(\mathbb{C}) \mid \det(M) \in H\}$ where $H = \det(G) \subset \mathbb{C}^\times$. In order to determine H, one can use the fact that $H = \det(G)$ is the difference Galois group of $\phi y = (\det A)y = by$ (this follows for instance from Tannakian duality [13, Section 1.4]).

Our first task, undertaken in the present section, is to study the reducibility of G. The imprimitivity of G will be considered in Section 5.

4.1. **Riccati equation and irreducibility.** The non linear difference equation

$$(\phi(u) + a)u = -b \tag{4.2}$$

is called the Riccati equation associated to equation (4.1). A straightforward calculation shows that u is a solution of this equation if and only if $\phi - u$ is a right factor of $\phi^2 + a\phi + b$, whence its link with irreducibility.

In what follows, we denote by I_2 the identity matrix of $\mathrm{GL}_2(\mathbb{C})$.

Lemma 4.1. *The following statements hold:*

1. *If (4.2) has one and only one solution in K then G is reducible but not completely reducible.*
2. *If (4.2) has exactly two solutions in K then G is completely reducible but not an algebraic subgroup of $\mathbb{C}^\times I_2$.*
3. *If (4.2) has at least three solutions in K then it has infinitely many solutions in K and G is an algebraic subgroup of $\mathbb{C}^\times I_2$.*
4. *If none of the previous cases occur then G is irreducible.*

Proof. The proof of this lemma is identical to that of [5, Theorem 4.2], to whom we refer for more details.

(1) We assume that (4.2) has one and only one solution $u \in K$. A straightforward calculation shows that

$$\phi(T)AT^{-1} = \begin{pmatrix} u & * \\ 0 & b/u \end{pmatrix} \qquad \text{for} \quad T := \begin{pmatrix} 1-u & 1 \\ -u & 1 \end{pmatrix} \in \mathrm{GL}_2(K).$$

We deduce from this and from Corollary 2.7 that G is reducible.

Moreover, if G was completely reducible then, in virtue of Corollary 2.7, $\phi(T)AT^{-1}$ would be diagonal for some $T := (t_{i,j})_{1\leq i,j\leq 2} \in \mathrm{GL}_2(K)$. Equating the entries of the antidiagonal of $\phi(T)AT^{-1}$ with 0, we find that $-\frac{t_{21}}{t_{22}}, -\frac{t_{11}}{t_{12}} \in K$ are solutions of the Riccati equation. Since $\det(T) \neq 0$, these solutions are distinct, whence a contradiction.

(2) Assume that (4.2) has exactly two solutions $u_1, u_2 \in K$. We have

$$\phi(T)AT^{-1} = \begin{pmatrix} u_1 & 0 \\ 0 & u_2 \end{pmatrix} \qquad \text{for} \quad T := \frac{1}{u_1-u_2}\begin{pmatrix} -u_2 & 1 \\ -u_1 & 1 \end{pmatrix} \in \mathrm{GL}_2(K).$$

We deduce from this and from Corollary 2.7 that G is completely reducible.

Moreover, if G was an algebraic subgroup of $\mathbb{C}^\times I_2$ then, according to Corollary 2.7, there would exist $u \in K$ and $T = (t_{i,j})_{1\leq i,j\leq 2} \in \mathrm{GL}_2(K)$ such that

$$\phi(T)AT^{-1} = uI_2.$$

This equality implies that t_{21} and t_{22} are non zero and that, for all $c, d \in \mathbb{C}$ with $ct_{2,2} + dt_{1,2} \neq 0$,

$$-\frac{ct_{21} + dt_{11}}{ct_{22} + dt_{12}} \in K$$

is a solution of (4.2). It is easily seen that we get in this way infinitely many solutions of the Riccati equation, this is a contradiction.

(3) Assume that (4.2) has at least three solutions $u_1, u_2, u_3 \in K$. The proof of assertion (2) of the present lemma shows that $\phi Y = AY$ is isomorphic over K to $\phi Y = \begin{pmatrix} u_i & 0 \\ 0 & u_j \end{pmatrix} Y$ for all $1 \leq i < j \leq 3$. Therefore, there exists $T \in \mathrm{GL}_2(K)$ such that

$$\phi(T)\begin{pmatrix} u_1 & 0 \\ 0 & u_2 \end{pmatrix} = \begin{pmatrix} u_1 & 0 \\ 0 & u_3 \end{pmatrix} T.$$

Equating the second columns in this equality, we see that there exists $f \in K^\times$ such that either $u_1 = \frac{\phi f}{f} u_2$ or $u_3 = \frac{\phi f}{f} u_2$; up to renumbering, one can assume that the former case holds true. It follows that $\phi Y = AY$ is isomorphic over K to

$$\phi Y = (u_1 I_2)Y$$

and, according to Corollary 2.7, G is an algebraic subgroup of $\mathbb{C}^\times I_2$. We have shown during the proof of statement (2) that this implies that the Riccati equation (4.2) has infinitely many solutions in K.

(4) Assume that G is reducible. According to Corollary 2.7, there exists $T = (t_{i,j})_{1\leq i,j\leq 2} \in \mathrm{GL}_2(K)$ such that $\phi(T)AT^{-1}$ is upper triangular. Then $t_{22} \neq 0$ and $-\frac{t_{21}}{t_{22}} \in K$ is a solution of the Riccati equation (4.2). This proves claim (4). □

In the proof of the previous lemma, we have shown the following result, which we state independently for ease of reference.

Lemma 4.2. *The following properties are equivalent:*

- *The Riccati equation* (4.2) *has at least three solutions in* K.
- *The Riccati equation* (4.2) *has infinitely many solutions in* K.
- *The difference Galois group* G *is a subgroup of* $\mathbb{C}^\times I_2$.
- *There exist* $u \in K^\times$ *and* $T \in \mathrm{GL}_2(K)$ *such that* $\phi(T)AT^{-1} = uI_2$.

We shall now state and prove one more lemma.

Lemma 4.3. *Let* $\Lambda'' \subset \Lambda'$ *be sublattices of* Λ *such that the quotient* Λ'/Λ'' *is cyclic. Assume that there exist* $u \in \mathrm{M}_{\Lambda''}^\times$ *and* $T \in \mathrm{GL}_2(\mathrm{M}_{\Lambda''})$ *such that*

$$\phi(T)AT^{-1} = uI_2. \tag{4.3}$$

Then, the Riccati equation (4.2) *has at least two distinct solutions in* $\mathrm{M}_{\Lambda'}$.

Proof. The Galois extension $\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'}$ is cyclic of order $k := [\mathrm{M}_{\Lambda''} : \mathrm{M}_{\Lambda'}]$. Its Galois group $\mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$ is generated by the field automorphism σ_1 given by $\sigma_1(f(z)) = f(z+\lambda')$, where $\lambda' \in \Lambda'$ is a representative of a generator of Λ'/Λ''. Note that the action of $\mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$ on $\mathrm{M}_{\Lambda''}$ commutes with the action of ϕ. Applying σ_1 to equation (4.3), we get

$$\phi(\sigma_1(T))A\sigma_1(T)^{-1} = \sigma_1(u)I_2,$$

so

$$\phi(S)u = \sigma_1(u)S, \qquad \text{with} \quad S := \sigma_1(T)T^{-1} \in \mathrm{GL}_2(\mathrm{M}_{\Lambda''}).$$

It follows that there exists $g_{\sigma_1} \in \mathrm{M}_{\Lambda''}^\times$ (namely, one of the non zero entries of S) such that

$$\sigma_1(u) = \frac{\phi(g_{\sigma_1})}{g_{\sigma_1}} u.$$

Consider the norm

$$\mathrm{N} := \mathrm{N}_{\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'}}(g_{\sigma_1}) = \prod_{\sigma \in \mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})} \sigma(g_{\sigma_1}) \in \mathrm{M}_{\Lambda'}^\times.$$

We have

$$\phi(\mathrm{N}) = \prod_{\sigma \in \mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})} \sigma\left(\frac{\sigma_1(u) g_{\sigma_1}}{u}\right) = \prod_{\sigma \in \mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})} \sigma(g_{\sigma_1}) = \mathrm{N},$$

so $\mathrm{N} = c \in (K^\phi)^\times = \mathbb{C}^\times$. Up to replacing g_{σ_1} by $g_{\sigma_1} c^{-1/k}$, we may assume that

$$\mathrm{N}_{\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'}}(g_{\sigma_1}) = 1.$$

Hilbert's 90 theorem [11, Section X.1] ensures that there exists $m \in \mathrm{M}_{\Lambda''}^\times$ such that

$$g_{\sigma_1} = \frac{m}{\sigma_1(m)}.$$

For any $\sigma = \sigma_1^j \in \mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$, we set

$$g_\sigma := g_{\sigma_1} \sigma_1(g_{\sigma_1}) \cdots \sigma_1^{j-1}(g_{\sigma_1}) = m/\sigma(m) \in \mathrm{M}_{\Lambda''}^\times;$$

we have

$$\sigma(u) = \frac{\phi(g_\sigma)}{g_\sigma} u.$$

It follows that

$$\widetilde{u} := \frac{\phi(m)}{m} u$$

is invariant under the action of $\mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$ and hence belongs to $\mathrm{M}_{\Lambda'}^{\times}$. We have

$$\phi(T')\, A\, (T')^{-1} = \widetilde{u} I_2, \qquad \text{with} \quad T' := mT \in \mathrm{GL}_2(\mathrm{M}_{\Lambda''}).$$

Applying $\sigma \in \mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$ to this equality, we get

$$\phi(\sigma(T'))\, A\, (\sigma(T'))^{-1} = \widetilde{u} I_2.$$

It follows that the matrix

$$C_\sigma := T' \sigma(T')^{-1} \in \mathrm{GL}_2(\mathrm{M}_{\Lambda''})$$

satisfies $\phi(C_\sigma) = C_\sigma$ and, hence, that its entries belong to $K^\phi = \mathbb{C}$. Moreover, $\sigma \mapsto C_\sigma$ is a 1-cocyle for the natural action of $\mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$ on $\mathrm{GL}_2(\mathbb{C})$ but this action is trivial so $\sigma \mapsto C_\sigma$ is a group morphism from $\mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$ to $\mathrm{GL}_2(\mathbb{C})$. Since $\mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$ is cyclic, this implies that there exists $P \in \mathrm{GL}_2(\mathbb{C})$ such that, for all $\sigma \in \mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$, the matrix $D_\sigma := P^{-1} C_\sigma^{-1} P \in \mathrm{GL}_2(\mathbb{C})$ is diagonal. We have, for all $\sigma \in \mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$,

$$\sigma(T'') = D_\sigma T'', \qquad \text{where} \quad T'' = (t''_{i,j})_{1 \le i,j \le 2} := P^{-1} T' \in \mathrm{GL}_2(\mathrm{M}_{k\Lambda}).$$

It follows that $u_1 := \frac{-t''_{11}}{t''_{12}}$ and $v_1 := \frac{-t''_{21}}{t''_{22}}$ are invariant by the action of $\mathrm{Gal}(\mathrm{M}_{\Lambda''} \,|\, \mathrm{M}_{\Lambda'})$ and hence belong to $\mathrm{M}_{\Lambda'}$. But u_1 and v_1 are solutions of the Riccati equation (4.2) (this was already used in the proof of assertion (2) of Lemma 4.1). So u_1 and v_1 are solutions in $\mathrm{M}_{\Lambda'}$ of the Riccati equation (4.2). □

We now come to the main result of this subsection.

Theorem 4.4. *The following statements hold:*

1. *The Galois group G is reducible if and only if the Riccati equation* (4.2) *has at least one solution in* $\mathrm{M}_{2\Lambda}$.
2. *The Galois group G is completely reducible if and only if the Riccati equation* (4.2) *has at least two solutions in* $\mathrm{M}_{2\Lambda}$.

Proof. In virtue of Lemma 4.1, it is sufficient to prove that:

(a) If the Riccati equation (4.2) has a unique solution in K, then it belongs to M_Λ.
(b) If the Riccati equation (4.2) has exactly two solutions in K, then they belong to $\mathrm{M}_{2\Lambda}$.
(c) If the Riccati equation (4.2) has at least three solutions in K, then the Riccati equation (4.2) has at least two solutions in $\mathrm{M}_{2\Lambda}$.

(a) Assume that the Riccati equation (4.2) has a unique solution u in K. Since $u(z)$, $u(z+1)$ and $u(z+\tau)$ are solutions of (4.2), we get

$$u(z) = u(z+1) = u(z+\tau)$$

and hence $u \in \mathrm{M}_\Lambda$.

(b) Assume that the Riccati equation (4.2) has exactly two solutions in K and let $u \in K$ be one of these solutions. Since $u(z)$, $u(z+1)$ and $u(z+2)$ are solutions of (4.2), we get $u(z+2) = u(z)$. Similarly, we have $u(z+2\tau) = u(z)$. So $u \in \mathrm{M}_{2\Lambda}$.

(c) What follows is inspired by [5, Theorem 4.2], but is a little bit subtler. Assume that the Riccati equation (4.2) has at least three solutions in K. According to Lemma 4.2, there exist $u \in K$ and $T = (t_{i,j})_{1\leq i,j\leq 2} \in \mathrm{GL}_2(K)$ such that

$$\phi(T)AT^{-1} = uI_2. \tag{4.4}$$

Let $k \in \mathbb{N}^*$ be such that the entries of T and u belong to $\mathrm{M}_{k\Lambda}$. Consider the following field extensions:

$$\mathrm{M}_\Lambda \subset L \subset \mathrm{M}_{k\Lambda}, \qquad \text{with} \quad L := \mathrm{M}_{\mathbb{Z}+k\tau\mathbb{Z}}\,.$$

Applying Lemma 4.3 to the extension $\mathrm{M}_{k\Lambda}\,|L$ and to the equation (4.4), we get that the Riccati equation (4.2) has two distinct solution u_1 and v_1 in L. If both of them belong to $\mathrm{M}_{2\Lambda}$ then the proof is completed. Otherwise, up to renumbering, we can assume that $u_1 \notin \mathrm{M}_{2\Lambda}$, i.e., that u_1 is not 2τ-periodic. Then

$$u_1(z), u_2(z) := u_1(z+\tau) \qquad \text{and} \qquad u_3(z) := u_1(z+2\tau)$$

are distinct solutions in L of the Riccati equation. For all integers $i, j \in \{1,2,3\}$ with $i < j$ we set $T_{i,j} := \frac{1}{u_i - u_j}\begin{pmatrix} -u_j & 1 \\ -u_i & 1 \end{pmatrix} \in \mathrm{GL}_2(L)$ and we have

$$\phi(T_{i,j})A(T_{i,j})^{-1} = \begin{pmatrix} u_i & 0 \\ 0 & u_j \end{pmatrix}$$

(this was already used in the proof of assertion (2) of Lemma 4.1). Therefore,

$$\phi\big(T_{1,3}(T_{1,2})^{-1}\big)\begin{pmatrix} u_1 & 0 \\ 0 & u_2 \end{pmatrix} = \begin{pmatrix} u_1 & 0 \\ 0 & u_3 \end{pmatrix} T_{1,3}(T_{1,2})^{-1}.$$

Equating the second columns in this equality, we see that there exists $f \in L^\times$ such that either $u_1 = \frac{\phi f}{f} u_2$ or $u_3 = \frac{\phi f}{f} u_2$; up to renumbering, we may assume that the former equality holds true. Then, we have

$$\phi\big(\widetilde{T}\big)A\widetilde{T}^{-1} = u_1 I_2 \tag{4.5}$$

with

$$u_1 \in L^\times \qquad \text{and} \qquad \widetilde{T} := \begin{pmatrix} 1 & 0 \\ 0 & f \end{pmatrix} T_{1,2} \in \mathrm{GL}_2(L).$$

Applying Lemma 4.3 to the extension $L|\,\mathrm{M}_\Lambda$ and to the equation (4.5), we see that the Riccati equation (4.2) has 2 distinct solutions in M_Λ. This concludes the proof. □

4.2. **On the solutions of the Riccati equation.** We refer to Section 3.1 for the notations (div_k, $\deg_k$, ω_k, etc.) used in this subsection. Let $k \geq 1$ be an integer. Consider $p_1 \in \Theta_k \cup \{0\}$ and $p_2, p_3 \in \Theta_k$ such that

$$a = \frac{p_1}{p_3} \qquad \text{and} \qquad b = \frac{p_2}{p_3}.$$

We let $u \in \mathrm{M}_{k\Lambda}$ be a potential solution of the Riccati equation (4.2).

Proposition 4.5. *We have*

$$u = \frac{\phi(r)}{r}\frac{p}{q}$$

for some $p, q, r \in \Theta_k$ such that

(*i*) $\mathrm{div}_k(p) \leq \mathrm{div}_k(p_2)$,
(*ii*) $\mathrm{div}_k(q) \leq \mathrm{div}_k(\phi^{-1}(p_3))$,
(*iii*) $\deg_k(p) = \deg_k(q)$,
(*iv*) $\omega_k(p/q) = \deg_k(r)h \mod k\Lambda$.

Proof. In what follows, the greatest common divisors (gcd) has to be understood in the ring $\mathcal{O}(\mathbb{C})$ of entire functions[3]. Let $p_4, p_5 \in \Theta_k$, with $\gcd(p_4, p_5) = 1$, be such that $u = p_4/p_5$. Let $r \in \Theta_k$ be a greatest common divisor of $\phi^{-1}(p_4)$ and p_5 and consider

$$p := \frac{p_4}{\phi(r)} \in \Theta_k \qquad \text{and} \qquad q := \frac{p_5}{r} \in \Theta_k.$$

By construction, we have

$$u = \frac{\phi r}{r}\frac{p}{q}$$

with $\gcd(p, \phi(q)) = \gcd(\phi(r)p, rq) = 1$. Then, the Riccati equation (4.2) becomes

$$p_3 \frac{\phi r}{r}\frac{p}{q}\phi\left(\frac{\phi r}{r}\frac{p}{q}\right) + p_1 \frac{\phi r}{r}\frac{p}{q} = -p_2,$$

i.e.,

$$p_3\phi^2(r)p\phi(p) + p_1\phi(r)p\phi(q) = -p_2 rq\phi(q).$$

It is now easily seen that p divides p_2 and that q divides $\phi^{-1}(p_3)$ in $\mathcal{O}(\mathbb{C})$. In terms of divisors, this is exactly (i) and (ii).

According to Lemma 3.3, we have

$$\frac{p}{q}(z + k\tau) = (-1)^{\deg_k(p/q)} e^{2i\pi\omega/k} e^{-2i\pi \deg_k(p/q) z/k} \frac{p}{q}(z)$$

for some representative ω of $\omega_k(p/q)$, and

$$\frac{\phi(r)}{r}(z + k\tau) = e^{-2i\pi \deg_k(r) h/k} \frac{\phi(r)}{r}(z).$$

Therefore

$$u(z + k\tau) = (-1)^{\deg_k(p/q)} e^{2i\pi\omega/k} e^{-2i\pi \deg_k(p/q) z/k} e^{-2i\pi \deg_k(r) h/k} u(z).$$

But $u \in \mathrm{M}_{k\Lambda}$, so $u(z + k\tau) = u(z)$ and, hence,

$$(-1)^{\deg_k(p/q)} e^{2i\pi\omega/k} e^{-2i\pi \deg_k(p/q) z/k} e^{-2i\pi \deg_k(r) h/k} = 1.$$

Hence $\deg_k(p/q) = 0$ and $\omega = \deg_k(r)h \mod k\Lambda$. This proves (iii) and (iv). □

5. Imprimitivity of the difference Galois group

We want to determine whether G is imprimitive, that is whether G is conjugate to a subgroup of

$$\left\{ \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix} \mid \alpha, \beta \in \mathbb{C}^\times \right\} \bigcup \left\{ \begin{pmatrix} 0 & \gamma \\ \delta & 0 \end{pmatrix} \mid \gamma, \delta \in \mathbb{C}^\times \right\}.$$

Theorem 5.1. *Assume that G is irreducible and that $a \neq 0$. Then, G is imprimitive if and only if there exists $u \in \mathrm{M}_{2\Lambda}$ such that*

$$\left(\phi^2(u) + \left(\phi^2\left(\frac{b}{a}\right) - \phi(a) + \frac{\phi(b)}{a}\right)\right) u = -\frac{\phi(b)b}{a^2}. \tag{5.1}$$

Proof. Arguing exactly as in [5, Theorem 4.6], we get that G is imprimitive if and only if equation (5.1) has a solution in K. But this is a Riccati-type equation, with ϕ replaced by ϕ^2. Therefore, the assertions (a), (b) and (c) given at the beginning of the proof of Theorem 4.4 allow us to conclude. □

Remark 5.2. If $a = 0$ then G is imprimitive in virtue of Corollary 2.7.

[3]According to [4], any finitely generated ideal of $\mathcal{O}(\mathbb{C})$ is principal, whence the existence of the greatest common divisor of any couple of elements of $\mathcal{O}(\mathbb{C})$. Such a greatest common divisor is unique up to multiplication by an unit of $\mathcal{O}(\mathbb{C})$.

Note that Proposition 4.5 can be used in order to find restrictions on the solutions of the above Riccati-type equation, but with ϕ replaced by ϕ^2.

6. Applications

We recall that $h \in \mathbb{C}$ is such that $h \mod \Lambda$ is not a torsion point of $\mathbb{C}/\Lambda$, i.e., that the corresponding point $\underline{h}$ of $\mathcal{E}(\mathbb{C})$ is not a torsion point.

6.1. **A discrete version of Lamé equation.** Let us consider the difference equation

$$\Delta_h^2 y = (A\wp(z) + B)y, \qquad \text{where} \quad \Delta_h y(z) = \frac{y(z+h) - y(z)}{h} \tag{6.1}$$

and $A, B \in \mathbb{C}$. This is a discrete version of the so-called Lamé differential equation

$$y''(z) = (A\wp(z) + B)y(z).$$

A straightforward calculation shows that equation (6.1) can be rewritten as follows:

$$\phi^2 y - 2\phi y + \big(-Ah^2\wp(z) - Bh^2 + 1\big)y = 0.$$

We will deduce Theorem 6.2 from the following theorem combined with a transcendence result due to Schneider.

Theorem 6.1. *Consider $a \in \mathbb{C}^\times$ and $b(z) = \alpha\wp(z) + \beta$ with $(\alpha, \beta) \in \mathbb{C}^\times \times \mathbb{C}$. Let $z_0 \in \mathbb{C}$ be such that $\wp(z_0) = -\beta/\alpha$.[4] If $\mathbb{Z}h \cap (\ell z_0 + \Lambda) = \{0\}$ for all $\ell \in \{-8, \dots, 8\}$ (this holds in particular if $\mathbb{Z}h \cap (\mathbb{Z}z_0 + \Lambda) = \{0\}$) then the difference Galois group over (K, ϕ) of $\phi^2 y + a\phi y + by = 0$ is $\mathrm{GL}_2(\mathbb{C})$.*

Proof of Theorem 6.1. For the notations, div_k, $[\cdot]_k$, etc, we refer to Section 3.1. Note that

$$\mathrm{div}_1(b) = [z_0]_1 + [-z_0]_1 - 2[0]_1.$$

So, we can write $a = \frac{p_1}{p_3}$ and $b = \frac{p_2}{p_3}$ for some $p_1, p_2, p_3 \in \Theta_1$ with

$$\mathrm{div}_1(p_2) = [z_0]_1 + [-z_0]_1$$

and

$$\mathrm{div}_1(p_3) = 2[0]_1.$$

We claim that G is irreducible, i.e., in virtue of Theorem 4.4, that the Riccati equation

$$(\phi(u) + a)u = -b \tag{6.2}$$

does not have any solution in $\mathrm{M}_{2\Lambda}$. Suppose to the contrary that it has a solution $u \in \mathrm{M}_{2\Lambda}$. Proposition 4.5 ensures that there exist $p, q, r \in \Theta_2$ such that

$$u = \frac{\phi(r)}{r}\frac{p}{q}$$

and

(i) $\mathrm{div}_2(p) \le \sum_{\ell_1,\ell_2 \in \{0,1\}} [\ell_1 + \ell_2\tau - z_0]_2 + [\ell_1 + \ell_2\tau + z_0]_2$,
(ii) $\mathrm{div}_2(q) \le \sum_{\ell_1,\ell_2 \in \{0,1\}} 2[\ell_1 + \ell_2\tau + h]_2$,
(iii) $\deg_2(p) = \deg_2(q)$,
(iv) $\omega_2(p/q) = \deg_2(r)h \mod 2\Lambda$.

[4] Any non constant elliptic function $f(z)$ has at least one zero (otherwise, $1/f(z)$ would be an entire elliptic function and hence would be constant). In particular, $\wp(z) + \beta/\alpha$ has a least one zero in $\mathbb{C}$.

Properties (i) and (ii) above imply that

$$\omega_2\left(p/q\right) = \ell z_0 - \deg_2(q)h \mod \Lambda$$

for some $\ell \in \{-4, \dots, 4\}$. We infer from this and from (iv) that

$$(\deg_2(r) + \deg_2(q))h = \ell z_0 \mod \Lambda.$$

The assumption on z_0 ensures that $\deg_2(r) = \deg_2(q) = 0$. It follows from (iii) that $\deg_2(p) = 0$ and hence u is a constant. But it is easily seen that equation (6.2) does not have any constant solution; this proves our claim.

We claim that G is not imprimitive, i.e., in virtue of Theorem 5.1, that

$$\left(\phi^2(u) + \frac{\phi^2(b)}{a} - a + \frac{\phi(b)}{a}\right) u = -\frac{\phi(b)b}{a^2} \tag{6.3}$$

does not have any solution in $\mathrm{M}_{2\Lambda}$. Suppose to the contrary that it has a solution $u \in \mathrm{M}_{2\Lambda}$. Equation (6.3) is of the form:

$$u\left(\phi^2(u) + \frac{p_1}{p_3}\right) = \frac{p_2}{p_3},$$

for some $p_1, p_2, p_3 \in \Theta_1$ with

$$\mathrm{div}_1(p_2) = 2[-2h]_1 + [z_0]_1 + [-z_0]_1 + [z_0 - h]_1 + [-z_0 - h]_1$$

and

$$\mathrm{div}_1(p_3) = 2[-2h]_1 + 2[-h]_1 + 2[0]_1.$$

We apply Proposition 4.5 with ϕ replaced by ϕ^2 to obtain the existence of $p, q, r \in \Theta_2$ such that

$$u = \frac{\phi^2(r)}{r}\frac{p}{q},$$

where

(v)
$$\mathrm{div}_2(p) \le \sum_{\ell_1,\ell_2\in\{0,1\}} 2[\ell_1 + \ell_2\tau - 2h]_2 + [\ell_1 + \ell_2\tau + z_0]_2 + [\ell_1 + \ell_2\tau - z_0]_2 + [\ell_1 + \ell_2\tau + z_0 - h]_2 + [\ell_1 + \ell_2\tau - z_0 - h]_2,$$

(vi) $\mathrm{div}_2(q) \le \sum_{\ell_1,\ell_2\in\{0,1\}} 2[\ell_1 + \ell_2\tau]_2 + 2[\ell_1 + \ell_2\tau + h]_2 + 2[\ell_1 + \ell_2\tau + 2h]_2,$

(vii) $\deg_2(p) = \deg_2(q),$

(viii) $\omega_2(p/q) = 2\deg_2(r)h \mod 2\Lambda.$

We claim that

(v′) $\mathrm{div}_2(p) \le \sum_{\ell_1,\ell_2\in\{0,1\}} [\ell_1 + \ell_2\tau + z_0]_2 + [\ell_1 + \ell_2\tau - z_0]_2,$

(vi′) $\mathrm{div}_2(q) \le \sum_{\ell_1,\ell_2\in\{0,1\}} 2[\ell_1 + \ell_2\tau]_2.$

Indeed, otherwise, arguing as for the proof of the irreducibility of G, we see that (v), (vi) and (viii) would lead to a relation of the form

$$(2\deg_2(r) + d)h = \ell z_0 \mod \Lambda$$

for some integer $\ell \in \{-8, \dots, 8\}$ and some integer $d > 0$ and this would contradict our assumption on z_0. Then, (viii) shows that

$$2\deg_2(r)h = \ell z_0 \mod \Lambda$$

for some integer $\ell \in \{-4, \dots, 4\}$ and hence $\deg_2(r) = 0$. Therefore, $u = p/q$ with $p, q \in \Theta_2$ satisfying (v′) and (vi′) above. Now remark that

$$\phi^2(u) + \frac{\phi^2(b)}{a} - a + \frac{\phi(b)}{a}$$

does not have poles in Λ. But any element of Λ is a pole of order 2 of the right hand side of equation (6.3), so any element of Λ is a pole of order at least 2 of u. It follows that (vi′) is an equality. Then, using (vii), we see that (v′) is also an equality.

So $\mathrm{div}_2(u) = \mathrm{div}_2(b)$ and hence $u = cb$ for some $c \in \mathbb{C}^\times$. We now plug $u = cb$ into equation (6.3) and we get:

$$c\left(\left(c + \frac{1}{a}\right)\phi^2(b) - a + \frac{\phi(b)}{a}\right) = -\frac{\phi(b)}{a^2}.$$

Since $-2h$ is a pole of $\phi^2(b)$ but not of $\phi(b)$, we get $c = -1/a$ and the above equation simplifies as follows:

$$\frac{-1}{a}\left(-a + \frac{\phi(b)}{a}\right) = -\frac{\phi(b)}{a^2}.$$

This gives $1 = 0$, whence a contradiction.

Therefore, G is irreducible and not imprimitive. So, as explained at the beginning of Section 4, $G = \{M \in \mathrm{GL}_2(\mathbb{C}) \,|\, \det(M) \in H\}$ where $H \subset \mathbb{C}^\times$ is the Galois group of $\phi y = by$, which is easily seen to be the multiplicative group $(\mathbb{C}^\times, \cdot)$. This concludes the proof. □

Theorem 6.2. *Assume that $\mathcal{E}$ is defined over $\overline{\mathbb{Q}}$ (i.e., $g_2, g_3 \in \overline{\mathbb{Q}}$) and that $h, A, B \in \overline{\mathbb{Q}}$ with $A \neq 0$. Then, the difference Galois group over (K, ϕ) of equation* (6.1) *is* $\mathrm{GL}_2(\mathbb{C})$.

Proof of Theorem 6.2. In virtue of Theorem 6.1, it is sufficient to prove that $\mathbb{Z}h \cap (\mathbb{Z}z_0 + \Lambda) = \{0\}$. Consider $m_1, m_2 \in \mathbb{Z}$ and $\lambda \in \Lambda$ such that $m_1 h = m_2 z_0 + \lambda$. We have $\wp(z_0) = \frac{-Bh^2+1}{Ah^2} \in \overline{\mathbb{Q}}$. It follows that either $m_2 z_0 + \lambda \in \Lambda$ or $\wp(m_2 z_0 + \lambda) \in \overline{\mathbb{Q}}$. (Indeed, suppose that $m_2 z_0 + \lambda \notin \Lambda$. Using equation (1.1), we see that $\wp'(z_0) \in \overline{\mathbb{Q}}$. Therefore, $\varphi(z_0)$ belongs to $\mathcal{E}(\overline{\mathbb{Q}})$, the map φ being defined in the introduction. Using the fact that φ is a group morphism and that $\mathcal{E}(\overline{\mathbb{Q}})$ is a subgroup of $\mathcal{E}(\mathbb{C})$, we get $\varphi(mz_0) \in \mathcal{E}(\overline{\mathbb{Q}})$. Therefore, $\wp(mz_0 + \lambda) = \wp(mz_0) \in \overline{\mathbb{Q}}$.) In the former case, we get $m_1 h \in \Lambda$ and hence $m_1 = 0$. In the later case, it follows from the work of Schneider [10] (for a reference in english, see Baker's book [1, Theorem 6.2]; see also Bertrand and Masser's papers [2, 7]) that $m_2 z_0 + \lambda$ and hence $m_1 h$ are transcendental numbers, which is excluded. □

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